Lecture 27: Vector-Valued Functions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define vector-valued functions and explain their use in representing curves in space.
- ☐ Evaluate limits and test for continuity of vector functions.
- ☐ Differentiate vector functions and interpret the derivative as the tangent vector to the curve.
- ☐ Find the velocity, speed, and acceleration of a particle moving along a curve in space.
- ☐ Apply differentiation rules for vector functions.
- ☐ Find the integrals of vector functions.

Vector Functions

scalar function: real-valued function (eg: exp, ln, polynomial, trig).

A vector function (or vector-valued function) is a function that takes variables and returns a vector.

$\underset{\sim}{r}$: real numbers $\longrightarrow$ vectors.

$$\boldsymbol{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}.$$

component functions.

Eg: $\underset{\sim}{r}(t) = (2t+1)\underset{\sim}{i} + t^2\underset{\sim}{j} + e^{2t}\underset{\sim}{k}$

$\underset{\sim}{r}(1) = 3\underset{\sim}{i} + \underset{\sim}{j} + e^2\underset{\sim}{k}$

real number; vector.

When a particle moves through space during a time interval I, we think of the particle's coordinates as functions defined on I

$$x = f(t), \quad y = g(t), \quad z = h(t), \quad t \in I.$$

Eg: If a particle move in a straight line.

Recall: Vector eqn of a line:

$$\underset{\sim}{r}(t) = \underset{\sim}{r}_0 + t\underset{\sim}{v}$$

the position of the particle at time t.

$$= \langle x_0, y_0, z_0 \rangle + t\langle v_1, v_2, v_3 \rangle$$

$$= \langle x_0 + tv_1, y_0 + tv_2, z_0 + tv_3 \rangle$$

$$= (x_0 + tv_1)\underset{\sim}{i} + (y_0 + tv_2)\underset{\sim}{j} + (z_0 + tv_3)\underset{\sim}{k}$$

component functions.

Vector Functions

In general, when considering the domain of a vector function, we take the common domain of its components, i.e., the set of all t's where all the component functions are defined.

Example: Determine the domain of the vector function

$$\boldsymbol{r}(t) = \langle e^t, \ln(3-t), \sqrt{t} \rangle.$$

Domain of e^t : $(-\infty, \infty)$

Domain of $\ln(3-t)$: $(-\infty, 3)$

Domain of $\sqrt{t}$: $[0, \infty)$

So Domain of $\boldsymbol{r}(t)$: $(-\infty, \infty) \cap (-\infty, 3) \cap [0, \infty) = [0, 3)$

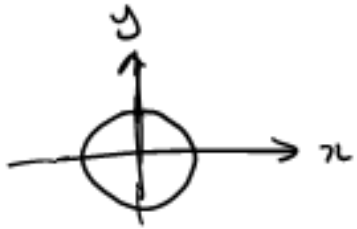
Vector Functions

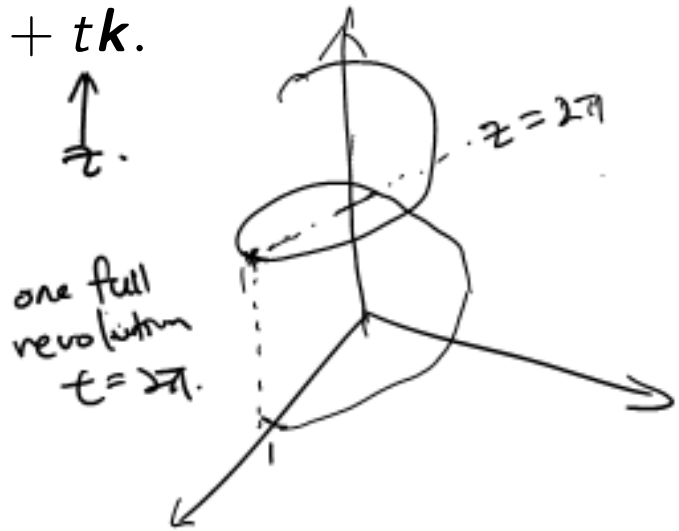
Example: Graph the vector function (helix)

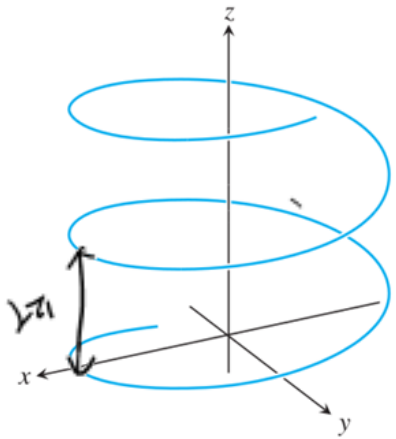
$$\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}.$$

$\cos^2 t + \sin^2 t = 1$

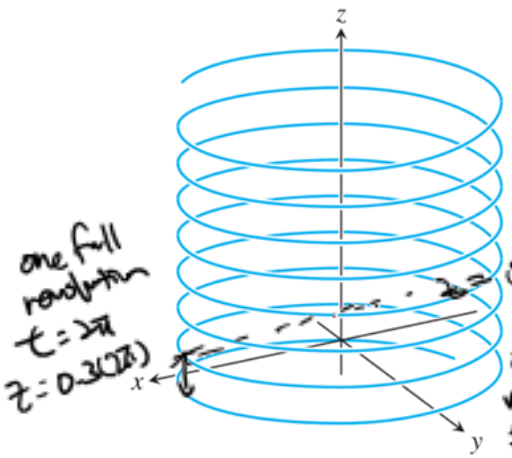
$x^2 + y^2 = 1$



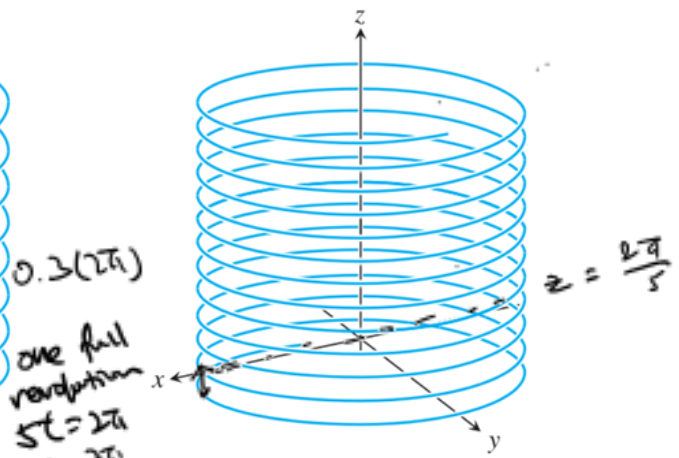




$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$



$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 0.3t\mathbf{k}$



$\mathbf{r}(t) = (\cos 5t)\mathbf{i} + (\sin 5t)\mathbf{j} + t\mathbf{k}$

Limits

Let $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ be a vector function with domain D, and let $\boldsymbol{L}$ be a vector. We say $\boldsymbol{r}$ has limit $\boldsymbol{L}$ as t approaches t_0 and write

$$\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{L}$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that, for all $t \in D$,

$$|\boldsymbol{r}(t) - \boldsymbol{L}| < \epsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

similar to the limit definition in real-valued function.

If $\boldsymbol{L} = L_1\boldsymbol{i} + L_2\boldsymbol{j} + L_3\boldsymbol{k}$, then $\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{L}$ if

$$\lim_{t \to t_0} f(t) = L_1, \quad \lim_{t \to t_0} g(t) = L_2, \quad \text{and} \quad \lim_{t \to t_0} h(t) = L_3.$$

$r(t) = \langle f(t), g(t), h(t) \rangle$

$\lim_{t \to t_0} r(t) = \langle \lim_{t \to t_0} f(t), \lim_{t \to t_0} g(t), \lim_{t \to t_0} h(t) \rangle$

Limits

Example: Find $\lim_{t \to \pi/4} \boldsymbol{r}(t)$ if $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}$.

$$\lim_{t \to \pi/4} \underset{\sim}{r}(t) = \left(\lim_{t \to \pi/4} \cos t \right) \underset{\sim}{i} + \left(\lim_{t \to \pi/4} \sin t \right) \underset{\sim}{j} + \left(\lim_{t \to \pi/4} t \right) \underset{\sim}{k}$$

$$= \frac{1}{\sqrt{2}} \underset{\sim}{i} + \frac{1}{\sqrt{2}} \underset{\sim}{j} + \frac{\pi}{4} \underset{\sim}{k}$$

Continuity

A vector function $\boldsymbol{r}(t)$ is continuous at a point $t = t_0$ in its domain if $\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{r}(t_0)$. The function is continuous if it is continuous at every point in its domain.

similar to the real-valued function
(f is cts at $t=t_0$ if $\lim_{t \to t_0} f(t) = f(t_0)$)

$\boldsymbol{r}(t)$ is continuous at $t = t_0$ if and only if each component function is continuous at $t = t_0$.

Example:

cts over $\mathbb{R}$ ↑ cts over $\mathbb{R}$ ↑ cts over $\mathbb{R}$.

The function $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}$ is continuous for all $t \in \mathbb{R}$.

The function $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + \lfloor t \rfloor \boldsymbol{k}$ is discontinuous at every integer.

↓ cts over $\mathbb{R}$ ↓ cts $\mathbb{R}$ ↓ not cts at every integer.

* $\underset{\sim}{r}(t)$ is cts over the intersection of the intervals where each of the component function is cts.

Derivatives and Motion

$= \langle f(t), g(t), h(t) \rangle$

Let $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ be the position vector of a particle moving along a curve in space.

Suppose that f, g, and h are differentiable functions of t.

The difference between the particle's positions at time t and the time $t + \Delta t$ is the vector

position at time $t+\Delta t$ ↓ ← position at time t

$$\Delta \boldsymbol{r} = \boldsymbol{r}(t + \Delta t) - \boldsymbol{r}(t)$$

$$= [f(t+\Delta t)\underset{\sim}{i} + g(t+\Delta t)\underset{\sim}{j} + h(t+\Delta t)\underset{\sim}{k}] - [f(t)\underset{\sim}{i} + g(t)\underset{\sim}{j} + h(t)\underset{\sim}{k}]$$

$$= [f(t+\Delta t) - f(t)]\underset{\sim}{i} + [g(t+\Delta t) - g(t)]\underset{\sim}{j} + [h(t+\Delta t) - h(t)]\underset{\sim}{k}$$

$$\underset{\sim}{r}'(t) = \lim_{\Delta t \to 0} \frac{\Delta \underset{\sim}{r}}{\Delta t} = \lim_{\Delta t \to 0} \left[\frac{f(t+\Delta t) - f(t)}{\Delta t} \right] \underset{\sim}{i}$$

$$+ \lim_{\Delta t \to 0} \left[\frac{g(t+\Delta t) - g(t)}{\Delta t} \right] \underset{\sim}{j}$$

$$+ \lim_{\Delta t \to 0} \left[\frac{h(t+\Delta t) - h(t)}{\Delta t} \right] \underset{\sim}{k}$$

$$= f'(t)\underset{\sim}{i} + g'(t)\underset{\sim}{j} + h'(t)\underset{\sim}{k}$$

$$\frac{d\underset{\sim}{r}}{dt} = \frac{df}{dt}\underset{\sim}{i} + \frac{dg}{dt}\underset{\sim}{j} + \frac{dh}{dt}\underset{\sim}{k}$$

$$r'(t) = \langle f'(t),\ g'(t),\ h'(t) \rangle$$

Derivatives and Motion

The vector function $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ has a derivative (is differentiable) at t if f, g, and h have derivatives at t. The derivative is the vector function

$$\boldsymbol{r}'(t) = \frac{d\boldsymbol{r}}{dt} = \lim_{\Delta t \to 0} \frac{\boldsymbol{r}(t+\Delta t) - \boldsymbol{r}(t)}{\Delta t} = \frac{df}{dt}\boldsymbol{i} + \frac{dg}{dt}\boldsymbol{j} + \frac{dh}{dt}\boldsymbol{k}.$$

Example: Find the derivative of $\boldsymbol{r}(t) = 3t^2\boldsymbol{i} + e^{2t}\boldsymbol{j} + \ln(t+1)\boldsymbol{k}$.

$$\underset{\sim}{r}'(t) = \frac{d(3t^2)}{dt}\underset{\sim}{i} + \frac{d(e^{2t})}{dt}\underset{\sim}{j} + \frac{d(\ln(t+1))}{dt}\underset{\sim}{k}$$

$$= 6t\,\underset{\sim}{i} + 2e^{2t}\,\underset{\sim}{j} + \frac{1}{t+1}\,\underset{\sim}{k}$$

Derivatives and Motion

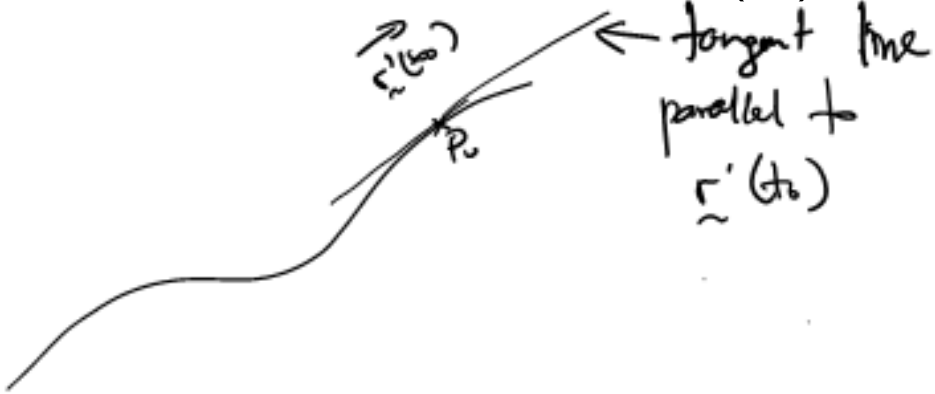
A vector function is differentiable if it is differentiable at every point of its domain.

$\frac{df}{dt}, \frac{dg}{dt}, \frac{dh}{dt}$ cannot be all zero at any point.

A curve $\boldsymbol{r}$ is smooth if $\frac{d\boldsymbol{r}}{dt}$ is continuous and never $\mathbf{0}$. Eg. helix.

On a smooth curve, if $\boldsymbol{r}(t)$ is the position vector of the point P, then $\boldsymbol{r}'(t)$ is the vector tangent to the curve at P. similar to real-valued function.

The tangent line to the curve at $P_0(f(t_0), g(t_0), h(t_0))$ is the line through the point P_0 parallel to the vector $\boldsymbol{r}'(t_0)$.



Eqn of the tangent line at B

$\underset{\sim}{S}(k) = \underset{\sim}{r}(t_0) + k\,\underset{\sim}{r}'(t_0)$

Derivatives and Motion

Example: Let $\boldsymbol{r}(t) = \langle \cos t, \sin t, t \rangle$ be a smooth curve. Find the equation of the tangent line at $t = \frac{\pi}{4}$.

At $t = \frac{\pi}{4}$, $\underset{\sim}{r}\left(\frac{\pi}{4}\right) = \left\langle \cos\frac{\pi}{4}, \sin\frac{\pi}{4}, \frac{\pi}{4} \right\rangle$

$$= \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{\pi}{4} \right\rangle$$

$$\underset{\sim}{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$$

At $t = \frac{\pi}{4}$, $\underset{\sim}{r}'\left(\frac{\pi}{4}\right) = \left\langle -\sin\frac{\pi}{4}, \cos\frac{\pi}{4}, 1 \right\rangle$

$$= \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1 \right\rangle$$

Equation of the tangent line, $\underset{\sim}{s}(k) = \underset{\sim}{r}\left(\frac{\pi}{4}\right) + k\,\underset{\sim}{r}'\left(\frac{\pi}{4}\right)$

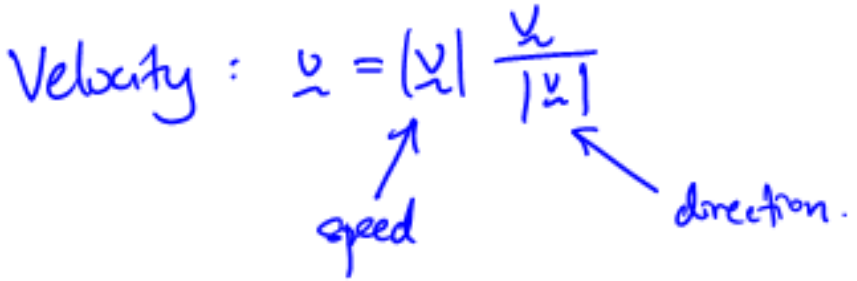
$$= \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{\pi}{4} \right\rangle + k\left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1 \right\rangle$$

Ex: Write it in parametric form and Cartesian form.

Derivatives and Motion

Let $\boldsymbol{r}$ be the position vector of a particle moving along a smooth curve in space.

1. Velocity is the derivative of position: $\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}$.
2. Speed is the magnitude of velocity: Speed $= |\boldsymbol{v}|$.
3. Acceleration is the derivative of velocity: $\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \frac{d^2\boldsymbol{r}}{dt^2}$.
4. The unit vector $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ is the direction of motion at time t.



Derivatives and Motion

Example: Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector

$$\boldsymbol{r}(t) = 2\cos t\,\boldsymbol{i} + 2\sin t\,\boldsymbol{j} + 5\cos^2 t\,\boldsymbol{k}.$$

Velocity: $\underset{\sim}{v}(t) = \underset{\sim}{r}'(t) = -2\sin t\,\underset{\sim}{i} + 2\cos t\,\underset{\sim}{j} + 10\cos t(-\sin t)\,\underset{\sim}{k}$

$\sin 2t = 2\cos t \sin t$

$= -2\sin t\,\underset{\sim}{i} + 2\cos t\,\underset{\sim}{j} - 5\sin 2t\,\underset{\sim}{k}$

Speed: $|\underset{\sim}{v}(t)| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + (-5\sin 2t)^2}$

$= \sqrt{4\sin^2 t + 4\cos^2 t + 25\sin^2 2t}$

$= \sqrt{4 + 25\sin^2 2t}$

Accerelation: $\underset{\sim}{a}(t) = \underset{\sim}{v}'(t) = -2\cos t\, \underset{\sim}{i} - 2\sin t\, \underset{\sim}{j} + (-5\cos 2t)(2)\underset{\sim}{k}$

$= -2\cos t\, \underset{\sim}{i} - 2\sin t\, \underset{\sim}{j} - 10\cos 2t\, \underset{\sim}{k}$

Example: Given that the position vector

$$\boldsymbol{r}(t) = 2t\cos 3t\,\boldsymbol{i} + \sin 3t\,\boldsymbol{j} + 5t\,\boldsymbol{k}.$$

Find the velocity of the particle after one full revolution.

velocity, $\underset{\sim}{v}(t) = \underset{\sim}{r}'(t) = [2\cos 3t + (2t)(-3\sin 3t)]\underset{\sim}{i}$

$+ 3\cos 3t\,\underset{\sim}{j} + 5\underset{\sim}{k}$

$= (2\cos 3t - 6t\sin 3t)\,\underset{\sim}{i}$

$+ 3\cos 3t\,\underset{\sim}{j} + 5\underset{\sim}{k}$

At one full revolution: $3t = 2\pi$ $(\theta = 2\pi)$

$t = \frac{2\pi}{3}$

$\underset{\sim}{v}\left(\frac{2\pi}{3}\right) = \left[2\cos 3\left(\frac{2\pi}{3}\right) - 6\left(\frac{2\pi}{3}\right)\sin 3\left(\frac{2\pi}{3}\right)\right]\underset{\sim}{i} + 3\cos 3\left(\frac{2\pi}{3}\right)\underset{\sim}{j} + 5\underset{\sim}{k}$

$= 2\underset{\sim}{i} + 3\underset{\sim}{j} + 5\underset{\sim}{k}$

Example: Given that the position vector of the particle is

$$\underset{\sim}{r}(t) = (2\cos t)\,\underset{\sim}{i} + (2\sin t)\,\underset{\sim}{j} + (5\cos^2 t)\,\underset{\sim}{k}$$

Find the times, if any, when the particle's velocity is orthogonal to its acceleration. find the t when $\underset{\sim}{v}(t)\cdot\underset{\sim}{a}(t)=0$.

$$\underset{\sim}{v}(t) = \underset{\sim}{r}'(t) = -2\sin t\,\underset{\sim}{i} + 2\cos t\,\underset{\sim}{j} - 5\sin 2t\,\underset{\sim}{k}$$

$$\underset{\sim}{a}(t) = \underset{\sim}{v}'(t) = -2\cos t\,\underset{\sim}{i} - 2\sin t\,\underset{\sim}{j} - 10\cos 2t\,\underset{\sim}{k}$$

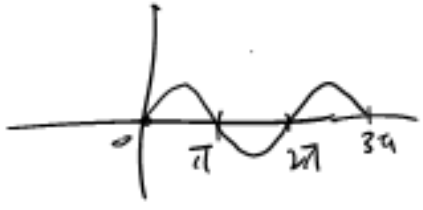
$$\underset{\sim}{v}\cdot\underset{\sim}{a} = 4\sin t\cos t - 4\cos t\sin t + 50\sin 2t\cos 2t = 0$$

$$25\sin 4t = 0$$

$$\sin 4t = 0$$

$$4t = n\pi,\ n\in\mathbb{Z}$$

$$t = \frac{n\pi}{4},\ n\in\mathbb{Z}.$$



Differentiation Rules

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be differentiable vector functions of t, $\boldsymbol{C}$ a constant vector, c any scalar, and f any differentiable scalar function. $\underset{\sim}{c} = \langle a_1, a_2, a_3 \rangle$

1. Constant Function Rule : $\frac{d}{dt}\boldsymbol{C} = \boldsymbol{0}$
2. Scalar Multiple Rules : $\frac{d}{dt}[c\boldsymbol{u}(t)] = c\boldsymbol{u}'(t)$
   $\frac{d}{dt}[f(t)\boldsymbol{u}(t)] = f'(t)\boldsymbol{u}(t) + f(t)\boldsymbol{u}'(t)$
3. Sum Rule : $\frac{d}{dt}[\boldsymbol{u}(t) + \boldsymbol{v}(t)] = \boldsymbol{u}'(t) + \boldsymbol{v}'(t)$
4. Difference Rule : $\frac{d}{dt}[\boldsymbol{u}(t) - \boldsymbol{v}(t)] = \boldsymbol{u}'(t) - \boldsymbol{v}'(t)$
5. Dot Product Rule : $\frac{d}{dt}[\boldsymbol{u}(t) \cdot \boldsymbol{v}(t)] = \boldsymbol{u}'(t) \cdot \boldsymbol{v}(t) + \boldsymbol{u}(t) \cdot \boldsymbol{v}'(t)$
6. Cross Product Rule : $\frac{d}{dt}[\boldsymbol{u}(t) \times \boldsymbol{v}(t)] = \boldsymbol{u}'(t) \times \boldsymbol{v}(t) + \boldsymbol{u}(t) \times \boldsymbol{v}'(t)$
7. Chain Rule : $\frac{d}{dt}[\boldsymbol{u}(f(t))] = f'(t)\boldsymbol{u}'(f(t))$

Constant function Rule:

Let $\underset{\sim}{c} = \langle a_1, a_2, a_3 \rangle$

$$\frac{d}{dt}\underset{\sim}{c} = \left\langle \frac{d}{dt}(a_1), \frac{d}{dt}(a_2), \frac{d}{dt}(a_3) \right\rangle$$

$$= \langle 0, 0, 0 \rangle = \underset{\sim}{0}$$

Scalar multiple Rule:

Let $\underset{\sim}{u}(t) = \langle u_1(t), u_2(t), u_3(t) \rangle$

$$c\,\underset{\sim}{u}(t) = \langle c u_1(t), c u_2(t), c u_3(t) \rangle$$

$$\frac{d}{dt}\left(c\,\underset{\sim}{u}(t)\right) = \left\langle \frac{d}{dt}(c u_1(t)), \frac{d}{dt}(c u_2(t)), \frac{d}{dt}(c u_3(t)) \right\rangle$$

$$= \langle c u_1'(t), c u_2'(t), c u_3'(t) \rangle$$

$$= c \langle u_1'(t), u_2'(t), u_3'(t) \rangle$$

$$= c\,\underset{\sim}{u}'(t)$$

$$f(t)\,\underline{u}(t) = f(t)\langle u_1(t), u_2(t), u_3(t)\rangle$$

$$= \langle f(t)u_1(t),\ f(t)u_2(t),\ f(t)u_3(t)\rangle$$

$$\frac{d}{dt}(f(t)u(t)) = \left\langle \frac{d}{dt}(f(t)u_1(t)),\ \frac{d}{dt}(f(t)u_2(t)),\ \frac{d}{dt}(f(t)u_3(t))\right\rangle$$

$$= \langle f'(t)u_1(t) + f(t)u_1'(t),\ f'(t)u_2(t) + f(t)u_2'(t),$$

$$f'(t)u_3(t) + f(t)u_3'(t)\rangle$$

$$= \langle f'(t)u_1(t), f'(t)u_2(t), f'(t)u_3(t)\rangle$$

$$+ \langle f(t)u_1'(t),\ f(t)u_2'(t),\ f(t)u_3'(t)\rangle$$

$$= f'(t)\,\underline{u}(t) + f(t)\,\underline{u}'(t)$$

Ex: Try to verify the rest.

Vector Functions of Constant Length

When a particle is moving on a sphere centered at the origin, the position vector has a constant length equal to the radius of the sphere.

The velocity vector $\frac{d\boldsymbol{r}}{dt}$, tangent to the path of motion, is tangent to the sphere and hence perpendicular to $\boldsymbol{r}$.

For a differentiable vector function of constant length, $\boldsymbol{r}(t)$ and $\boldsymbol{r}'(t)$ are orthogonal.

$|\underset{\sim}{r}(t)| = c$, c is a constant.

$\underset{\sim}{r}(t) \cdot \underset{\sim}{r}(t) = |\underset{\sim}{r}(t)|^2 = c^2$

$\frac{d}{dt}\left[\underset{\sim}{r}(t) \cdot \underset{\sim}{r}(t)\right] = \frac{d}{dt}(c^2) = 0$

$\underset{\sim}{r}'(t) \cdot \underset{\sim}{r}(t) + \underset{\sim}{r}(t) \cdot \underset{\sim}{r}'(t) = 0$

$2(\underset{\sim}{r}'(t) \cdot \underset{\sim}{r}(t)) = 0$

$\underset{\sim}{r}'(t) \cdot \underset{\sim}{r}(t) = 0$

$\underset{\sim}{r}'(t) \perp \underset{\sim}{r}(t)$

Vector Functions of Constant Length

If $\boldsymbol{r}$ is a differentiable vector function of t and the length of $\boldsymbol{r}(t)$ is constant, then

$$\boldsymbol{r} \cdot \frac{d\boldsymbol{r}}{dt} = 0.$$

Example: $\boldsymbol{r}(t) = \cos t\, \boldsymbol{i} + \sin t\, \boldsymbol{j} + \boldsymbol{k}$

$$|\underline{r}(t)| = \sqrt{\cos^2 t + \sin^2 t + 1^2}$$

$$= \sqrt{2} \quad (\text{constant})$$

$$\underline{r}'(t) = -\sin(t)\,\underline{i} + \cos(t)\,\underline{j}$$

$$\underline{r}(t) \cdot \underline{r}'(t) = (\cos t)(-\sin t) + (\sin t)(\cos t) + (1)(0) = 0$$

Integrals of Vector Functions

A differentiable vector function $\boldsymbol{R}(t)$ is an antiderivative of a vector function $\boldsymbol{r}(t)$ on an interval I if $\frac{d\boldsymbol{R}}{dt} = \boldsymbol{r}$ at each point of I.

If $\boldsymbol{R}$ is an antiderivative of $\boldsymbol{r}$ on I, then every antiderivative of $\boldsymbol{r}$ on I has the form $\boldsymbol{R} + \boldsymbol{C}$ for some constant vector $\boldsymbol{C}$.

The indefinite integral of $\boldsymbol{r}$ with respect to t is the set of all antiderivatives of $\boldsymbol{r}$, denoted by

$$\int \boldsymbol{r}(t)\, dt.$$

$\underline{r}(t) = \langle f(t), g(t), h(t) \rangle$

$\int \underline{r}(t)\, dt = \langle \int f(t)\, dt, \int g(t)\, dt, \int h(t)\, dt \rangle + \underset{\sim}{C}$

Example: $\int ((\cos t)\boldsymbol{i} + \boldsymbol{j} - 2t\boldsymbol{k})\, dt$

$= \left(\int \cos t\, dt\right)\underline{i} + \left(\int 1\, dt\right)\underline{j} + \left(\int -2t\, dt\right)\underline{k} + \underset{\sim}{C}$

$= \sin t\, \underline{i} + t\, \underline{j} - t^2\, \underline{k} + \underset{\sim}{C}$

$= \int f(t)\, dt\, \underline{i} + \int g(t)\, dt\, \underline{j} + \int h(t)\, dt\, \underline{k} + \underset{\sim}{C}$

Integrals of Vector Functions

If the components of $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ are integrable over $[a, b]$, then so is $\boldsymbol{r}$, and the definite integral of $\boldsymbol{r}$ from a to b is

$$\int_a^b \boldsymbol{r}(t)\,dt = \left(\int_a^b f(t)\,dt\right)\boldsymbol{i} + \left(\int_a^b g(t)\,dt\right)\boldsymbol{j} + \left(\int_a^b h(t)\,dt\right)\boldsymbol{k}.$$

Example: $\displaystyle\int_0^{\pi} ((\cos t)\boldsymbol{i} + \boldsymbol{j} - 2t\boldsymbol{k})\,dt = \left(\int_0^{\pi} \cos t\,dt\right)\underset{\sim}{i} + \left(\int_0^{\pi} 1\,dt\right)\underset{\sim}{j} + \left(\int_0^{\pi} -2t\,dt\right)\underset{\sim}{k}$

$= [\sin t]_0^{\pi}\,\underset{\sim}{i} + [t]_0^{\pi}\,\underset{\sim}{j} + [-t^2]_0^{\pi}\,\underset{\sim}{k}$

$= \pi\,\underset{\sim}{j} - \pi^2\,\underset{\sim}{k}$

The Fundamental Theorem of Calculus for continuous vector functions:

$$\int_a^b \boldsymbol{r}(t)\,dt = \boldsymbol{R}(t)\Big]_a^b = \boldsymbol{R}(b) - \boldsymbol{R}(a).$$

Similar to real-valued function.

Integrals of Vector Functions

Example: A moving particle departed from the point $(4, 0, 0)$ with initial velocity $\boldsymbol{v}(0) = 3\boldsymbol{j}$. Suppose that its acceleration is $\boldsymbol{a}(t) = -(3\cos t)\boldsymbol{i} - (3\sin t)\boldsymbol{j} + 2\boldsymbol{k}$. Find its velocity and position at time t.

$$\underset{\sim}{a}(t) = -(3\cos t)\,\underset{\sim}{i} - (3\sin t)\,\underset{\sim}{j} + 2\underset{\sim}{k}$$

Velocity, $\underset{\sim}{v}(t) = \int \underset{\sim}{a}(t)\,dt = \left(\int -3\cos t\,dt\right)\underset{\sim}{i} + \left(\int -3\sin t\,dt\right)\underset{\sim}{j} + \left(\int 2\,dt\right)\underset{\sim}{k}$

$$+ \underset{\sim}{C}$$

$$= -3\sin t\,\underset{\sim}{i} + 3\cos t\,\underset{\sim}{j} + 2t\,\underset{\sim}{k} + \underset{\sim}{C}$$

$$\underset{\sim}{v}(0) = (-3\sin 0)\,\underset{\sim}{i} + (3\cos 0)\,\underset{\sim}{j} + 2(0)\,\underset{\sim}{k} + \underset{\sim}{C} = 3\underset{\sim}{j}$$

$$3\underset{\sim}{j} + \underset{\sim}{C} = 3\underset{\sim}{j}$$

$$\underset{\sim}{C} = \underset{\sim}{0}$$

$$\underset{\sim}{v}(t) = -3\sin t\,\underset{\sim}{i} + 3\cos t\,\underset{\sim}{j} + 2t\,\underset{\sim}{k}$$

Position, $\underset{\sim}{r}(t) = \int \underset{\sim}{v}(t)\,dt$

$$= \left(\int -3\sin t\,dt\right)\underset{\sim}{i} + \left(\int 3\cos t\,dt\right)\underset{\sim}{j} + \left(\int 2t\,dt\right)\underset{\sim}{k} + \underset{\sim}{D}$$

$$= 3\cos t\,\underset{\sim}{i} + 3\sin t\,\underset{\sim}{j} + t^2\,\underset{\sim}{k} + \underset{\sim}{D}$$

$$\underset{\sim}{r}(0) = \langle 4, 0, 0\rangle = 4\underset{\sim}{i}$$

$$3\cos 0\,\underset{\sim}{i} + 3\sin 0\,\underset{\sim}{j} + (0)^2\,\underset{\sim}{k} + \underset{\sim}{D} = 4\underset{\sim}{i}$$

$$3\underset{\sim}{i} + \underset{\sim}{D} = 4\underset{\sim}{i}$$

$$\underset{\sim}{D} = \underset{\sim}{i}$$

So, $\underset{\sim}{r}(t) = 3\cos t\,\underset{\sim}{i} + 3\sin t\,\underset{\sim}{j} + t^2\,\underset{\sim}{k} + \underset{\sim}{i}$

$$= (1 + 3\cos t)\,\underset{\sim}{i} + 3\sin t\,\underset{\sim}{j} + t^2\,\underset{\sim}{k}$$